

Math Class 11 Federal Board
Chapter # 8 & 9 Trigonometry
MOST IMPORTANT Questions

76
 $\downarrow$
 $\uparrow$
 $\times 2x$ } $\Rightarrow$

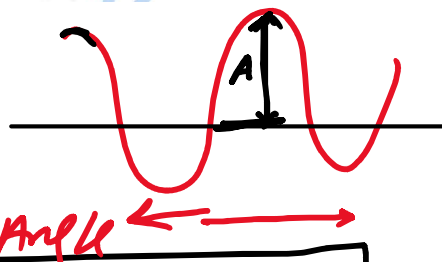
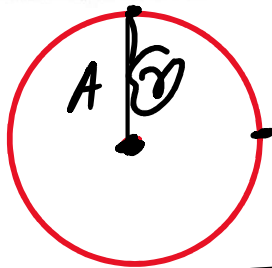
i) Vectors
 3 7

MCQs

Example 10:

A Ferris wheel has a diameter of 30 m with a center 17 m above the ground. It makes one complete revolution every 60 seconds.

- (1) Graph one complete period of the Ferris wheel that models the height in relation to time.
 Assume a rider starts at the lowest point.
- (2) Find the equation of the graph using the cosine function.
- (3) What is the height of the rider at 45 seconds?
- (4) At what time or times, the rider is at a height of 15 m?



$$y = A \cos(Bx + C) + D$$

$A = \text{Amplitude} = |A|$

$C = 0^\circ$
 (Note: $C = 0^\circ$ is circled in red)

$$|B| = \frac{2\pi \text{ rad}}{T \text{ s}}$$

$T = \text{Time Period}$

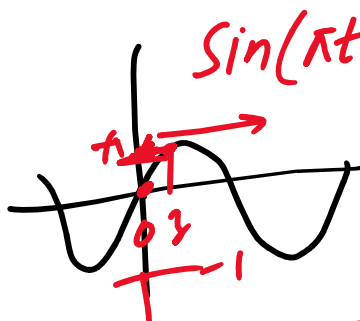
$$|B| = \frac{360^\circ}{T} \Rightarrow T = \frac{2\pi}{|B|}$$

$$T = \frac{360}{|B|}$$

$$y = A \cos(Bx) + C$$

$$y = A \cos(Bx) + C$$

$C \rightarrow$ Vertical ~~vertical~~ shift



$$\sin(\pi t) \Rightarrow \sin(\pi t + \pi/6)$$

$$\sin(\pi t) + 3$$



$$y = A \cos(Bx + C) + D$$

$A =$ Amplitude

$$= \frac{\text{Max value} - \text{Min value}}{2}$$

$D =$ Vertical shift

$$= \frac{\text{Max} + \text{Min}}{2}$$

$$|B| = \frac{2\pi}{T} \quad \text{or} \quad \frac{360^\circ}{T}$$

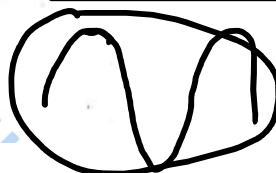
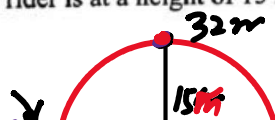
Example 10:

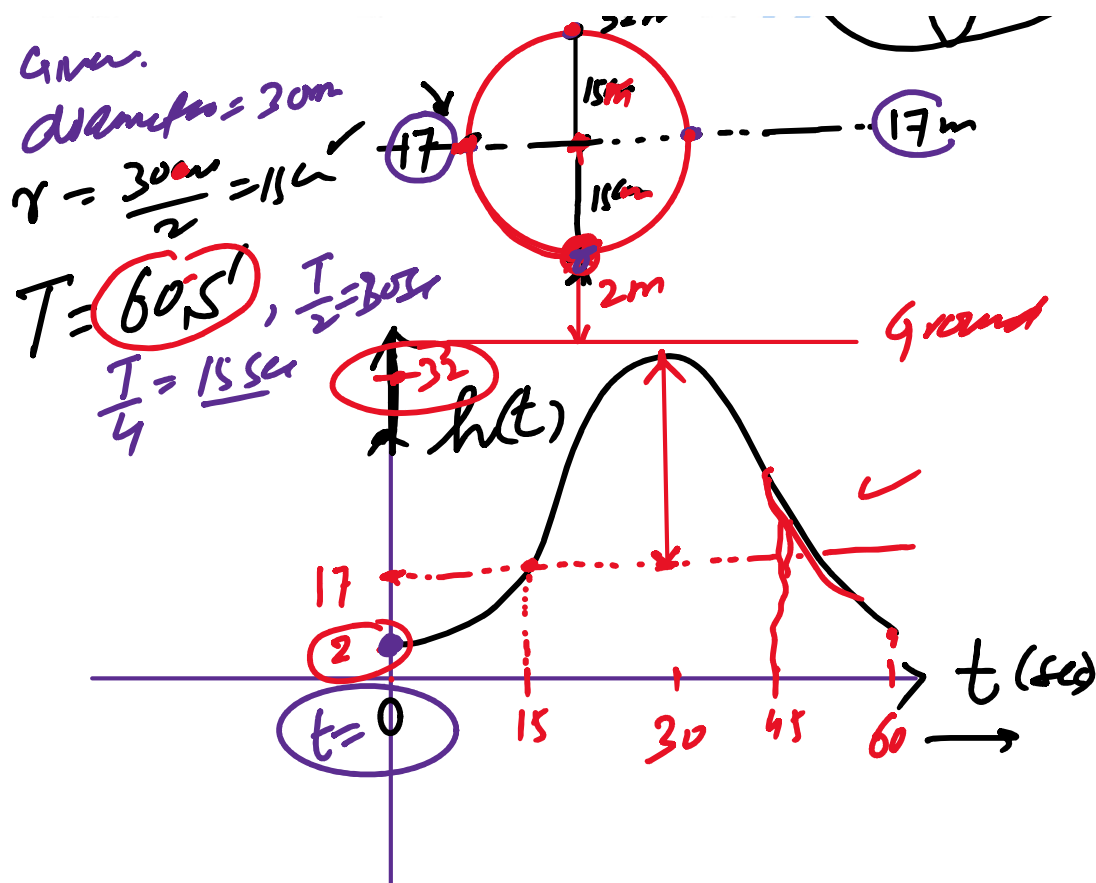
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- (1) Graph one complete period of the Ferris wheel that models the height in relation to time.
- (2) Assume a rider starts at the lowest point.
- (3) Find the equation of the graph using the cosine function.
- (4) What is the height of the rider at 45 seconds?
- (5) At what time or times, the rider is at a height of 15 m?

Given:

$$\text{Diameter} = 30 \text{ m}$$





Example 10:

A Ferris wheel has a diameter of 30 m with a center 17 m above the ground. It makes one complete revolution every 60 seconds.

- (1) Graph one complete period of the Ferris wheel that models the height in relation to time. Assume a rider starts at the lowest point.
- (2) Find the equation of the graph using the cosine function.
- (3) What is the height of the rider at 45 seconds?
- (4) At what time or times, the rider is at a height of 15 m?

(2) We have a sinusoidal fn

$$y = A \cos(Bx + C) + D$$

$$h(t) = A \cos(Bt) + C \rightarrow \textcircled{1}$$

Amplitude
 $= A = \frac{32 - 2}{2} = \frac{30}{2} = 15$

$$A = 15m$$

$$D = 2\pi = 2\pi = \frac{\pi}{1}$$

$$B = \frac{2\pi}{T} = \frac{2\pi}{60} = \frac{\pi}{30}$$

$$\boxed{B = \frac{\pi}{30}}$$

Vertical shift = $C = \frac{32+2}{2} = 17$

$$\boxed{C = 17}$$

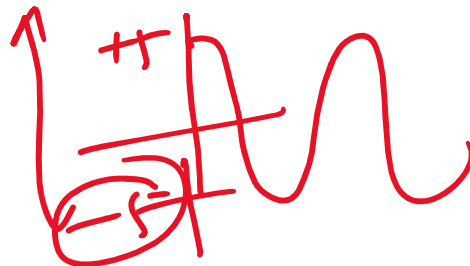
* $\Rightarrow$

$$\textcircled{1} \Rightarrow \boxed{h(t) = -15 \cos\left(\frac{\pi}{30}t\right) + 17}$$

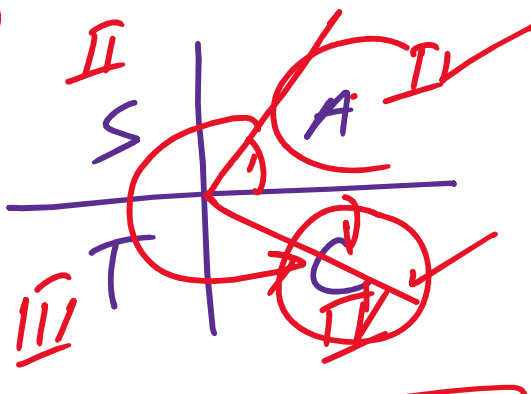
Test is best for the rest

$t \Rightarrow h(0) = -15(1) + 17 = 2$

$$y = -5 \cos(2t)$$



$\theta = 1.437$
True



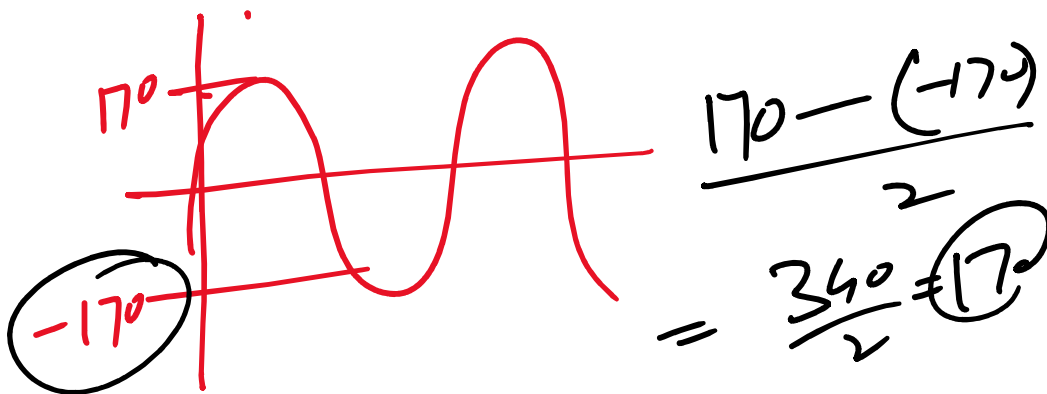
Chap 7.8

III · I (IV)

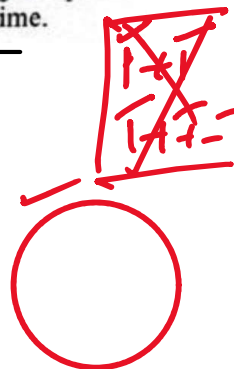
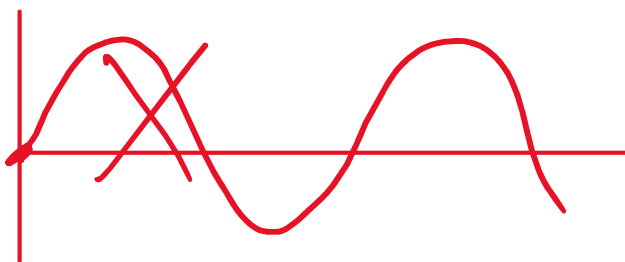
$$\boxed{\theta_2 = 2\pi - \theta_1}$$

3rd
2nd

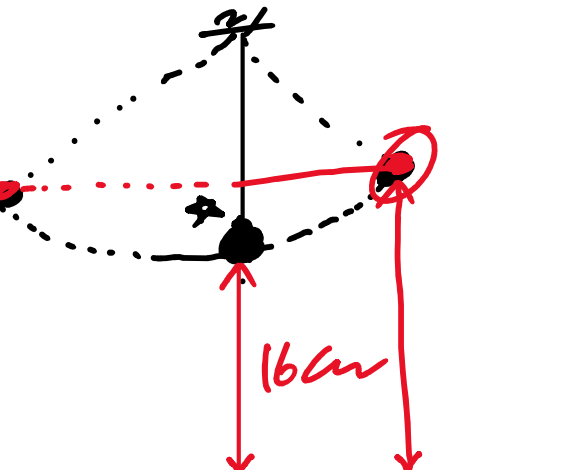
$\pi + \alpha$
 $\pi - \alpha$

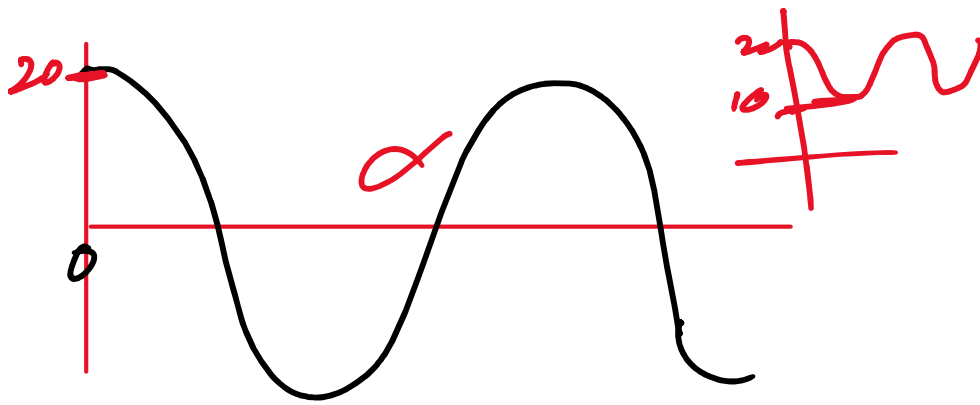


The pendulum of a clock swings with a periodic motion that can be represented by a trigonometric function. At rest, the pendulum is 16 cm above the base. The highest point of the swing is 20 cm above the base, and it takes 2 seconds for the pendulum to swing back and forth to its starting point once. Assume the pendulum is released from its highest point. Write a cosine equation that models the height of the pendulum as a function of time.



$T = 2\pi k$





$$\underline{\text{Sol}} \quad h(t) = A \cos(Bt) + C \quad \rightarrow \textcircled{1}$$

$$T = 2 \text{ sec}$$

$$A = \frac{\text{Max} - \text{Min}}{2} = \frac{20 - 16}{2} = \frac{4}{2} = 2$$

$$\boxed{A = 2}$$

$$B = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$$

$$\boxed{B = \pi}$$

$$C = \text{Vertical Shift} = \frac{\text{Max} + \text{Min}}{2}$$

$$= \frac{20 + 16}{2} = \frac{36}{2} = 18$$

$$\boxed{C = 18}$$

$$\textcircled{1} \Rightarrow \boxed{h(t) = 2 \cos(\pi t) + 18} \quad \checkmark$$

4. Find the periods of the following:

Example

(i) $y = 6 \sec(2x - 3)$ (ii) $y = \cos(5x + 4)$ (iii) $y = \cot 4x + \sin \frac{5x}{2}$

(iv) $y = 7 \sin(3x + 3)$ (v) $y = 5 \sin(2x + 3)$ (vi) $y = 2 \tan 3x + 7 \cos 5x$

a) Period = $\frac{2\pi}{2} = \pi$ ✓ $\frac{\pi}{3}, \frac{2\pi}{5}$
 b) = $\frac{2\pi}{5}$ ✓ = $\frac{2\pi}{1} = 2\pi$

(iii) $y = \cot 4x + \sin \frac{5x}{2}$

a) Period of $\cot 4x = T_1 = \frac{\pi}{4} \Rightarrow T_1 = \frac{\pi}{4}$

b) Period of $\sin \frac{5x}{2} = T_2 = \frac{2\pi}{\frac{5}{2}} = 2\pi \times \frac{2}{5} = \frac{4\pi}{5}$

$(\pi, 4)$ $T_2 = \frac{4\pi}{5}$

Thus $T = \text{Period of fn} = \frac{\text{LCM}(\pi, 4\pi)}{\text{HCF}(4, 1)}$

$\downarrow$ ✓ $\sin \left(\frac{4\pi}{1} \right) = 4\pi$

Trig. Functions	Domain	Range
$y = \sin x$	$\mathbb{R} = (-\infty, \infty)$ ✓	$-1 \leq y \leq 1$
$y = \cos x$	$\mathbb{R} = (-\infty, \infty)$ ✓	$-1 \leq y \leq 1$
$y = \tan x$	$\{x : x \in \mathbb{R} \wedge x \neq (2n+1)\frac{\pi}{2}, n \text{ an integer}\}$	$\mathbb{R} = (-\infty, \infty)$
$y = \cot x$	$\{x : x \in \mathbb{R} \wedge x \neq n\pi, n \text{ an integer}\}$	$\mathbb{R} = (-\infty, \infty)$
$y = \sec x$	$\{x : x \in \mathbb{R} \wedge x \neq (2n+1)\frac{\pi}{2}, n \text{ an integer}\}$	$1 \leq y \text{ and } y \leq -1$

Trig. Functions	Domain	Range
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$y = \cot x$	$\{x : x \in \mathbb{R} \wedge x \neq n\pi, n \text{ an integer}\}$	$\mathbb{R} = (-\infty, \infty)$
$y = \sec x$	$\{x : x \in \mathbb{R} \wedge x \neq (2n+1)\frac{\pi}{2}, n \text{ an integer}\}$	$1 \leq y \text{ and } y \leq -1$
$y = \operatorname{cosec} x$	$\{x : x \in \mathbb{R} \wedge x \neq n\pi, n \text{ an integer}\}$	$1 \leq y \text{ and } y \leq -1$

$y = \sin x$, am Aufg $x \rightarrow \boxed{\sin(x)} \rightarrow y$ ✓
 $D_y = \mathbb{R} \text{ or } (-\infty, \infty)$ ✓
 Range $\Rightarrow R_y = \boxed{[-1, 1]}$ ✓ $\rightarrow$ Check
 $\times \boxed{[-1, 1]} \rightarrow \text{open}$
 $(-1, 1)$

f. g. $\{y \mid y \in \mathbb{R} \wedge \underline{-1 \leq y \leq 1}\}$
 $\rightarrow -2 \sinh x$
 $\boxed{-2 \leq \sinh x \leq 2}$

$$\left(\frac{3}{2}\right) \cos$$

$$R_y \in \left[-\frac{3}{2}, \frac{3}{2}\right]$$

$$y = \tan \alpha = \frac{\sin \alpha}{\cos \alpha}$$

$$\cos \alpha \neq 0$$

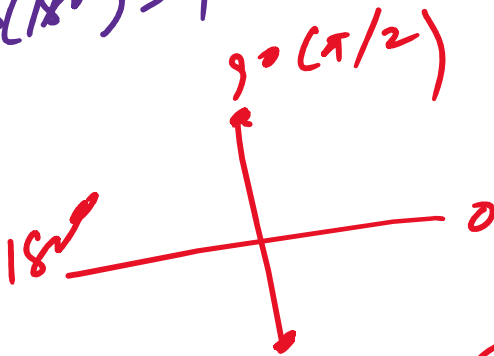
$$90^\circ \times \frac{\pi}{180}$$

$$\cos 90^\circ = 0$$

$$\cos(180^\circ) = -1$$

$$\cos \alpha = 0$$

$$\alpha = \cos^{-1}(0) = \frac{\pi}{2}$$



$$1 - \frac{\pi}{2}, \quad \frac{3\pi}{2}$$

$$3 \text{ or } 3\pi/2 \quad n=1 \quad 2n-1=1 \quad 90^\circ + 270^\circ$$

$$\tan \alpha$$

$$D_y = \alpha \in \mathbb{R} \wedge \alpha \neq (2n+1)\left(\frac{\pi}{2}\right)$$

$$1, 3, 5, 7, \dots$$

$$A.P. \{1\}$$

$$a = 1, \quad d = 2$$

$$a_n = a + (n-1)d$$

$$= 1 + (n-1)2$$

$$= 1 + 2n - 2$$

$$1, 3, 5, \dots = \underline{\underline{(2n+1)}}$$

$$\cot x = \frac{\cos x}{\sin x}$$

$$0, \pi, 2\pi$$

$$x \neq n\pi, n \in \mathbb{Z}$$

$y = \sec x$	$\{x : x \in \mathbb{R} \wedge x \neq (2n+1)\frac{\pi}{2}, n \text{ an integer}\}$	$1 \leq y \text{ and } y \leq -1$
$y = \csc x$	$\{x : x \in \mathbb{R} \wedge x \neq n\pi, n \text{ an integer}\}$	$1 \leq y \text{ and } y \leq -1$

$$y = \sec x = \frac{1}{\cos x}$$

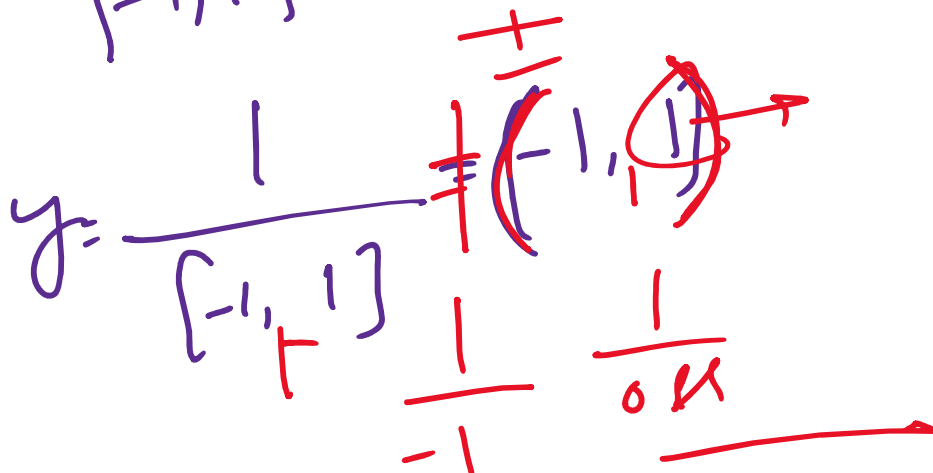
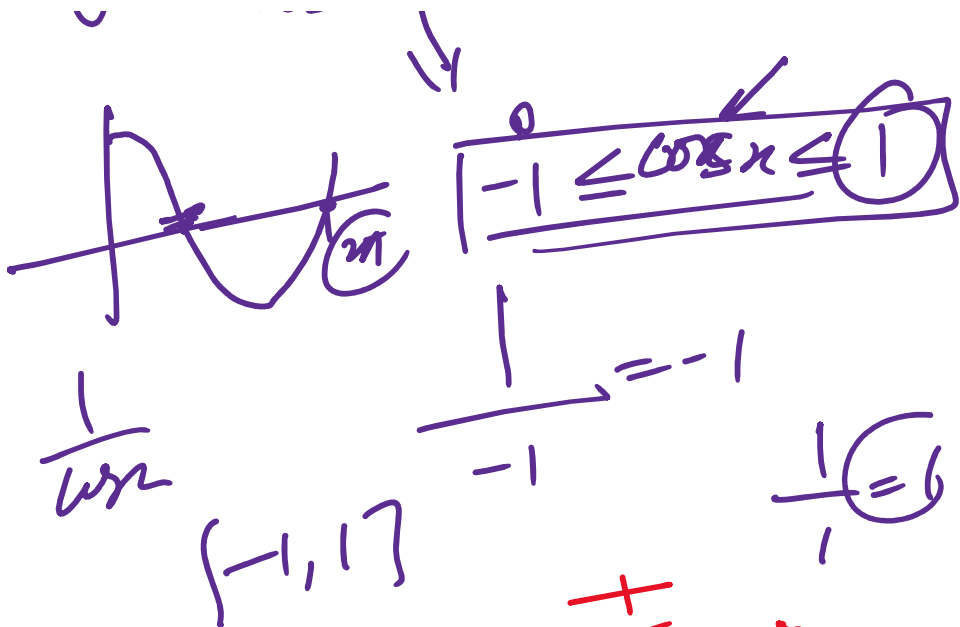
$\frac{1}{0} \rightarrow \infty$
 $\cos x \rightarrow \left((2n+1)\left(\frac{\pi}{2}\right) \right)$

$y = \csc x = \frac{1}{\sin x}$

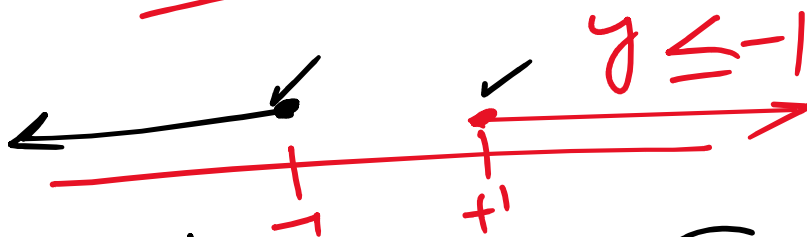
کی یہ کہ قتل کے لغو اس نے فعل سے جواب
 لے اس زور پر یہاں کا لکھا ہے

فہم

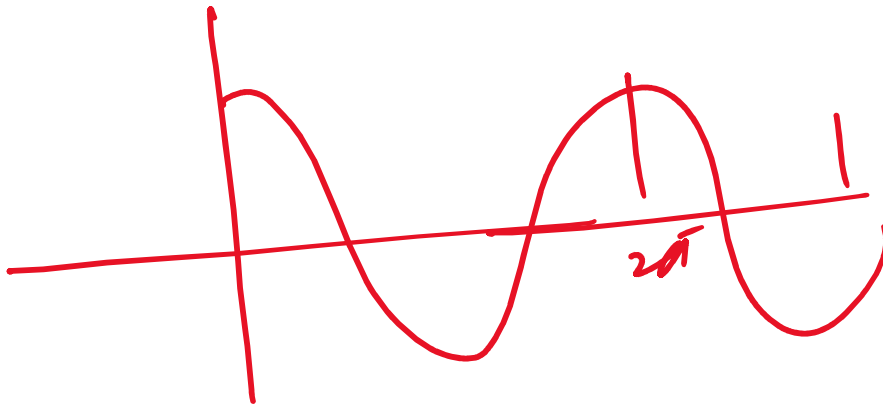
$$y = \frac{1}{\cos x}$$



$R_{\text{seen}} = y \geq 1$



$\frac{1}{\cos(\frac{x}{2})}$
 $\frac{x}{2} = 10$
 $x = 180$
 $\frac{1}{2}$
 $\frac{2\pi}{2}$

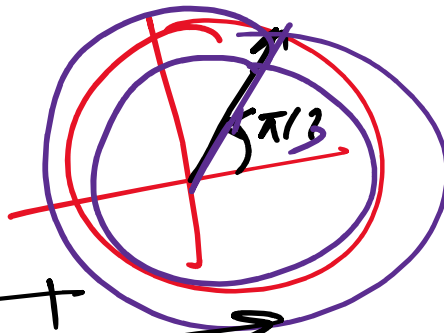


2π is the $\dots$
 $\dots 4\pi - 2\pi$

$$(2n\pi), n \in \mathbb{Z}$$

$$\frac{\pi}{3} + 2n\pi$$

$$\frac{\pi}{3} + (2n\pi)$$



$$-3 \leq 2 \cos x - 1 \leq 1$$

$$1 \leq \frac{1}{2 \cos x - 1} \leq \frac{-1}{3}$$

$$1 \leq y \leq \frac{-1}{3}$$

$$y \leq \frac{-1}{3} \text{ and } 1 \leq y$$

$$\text{e of } y = R_y = (-\infty, -\frac{1}{3}] \cup [1, +\infty)$$

$$5 < 10$$

$$\frac{1}{5} > \frac{1}{10}$$

$$0.2 > 0.1$$

$$-1 \leq \frac{1}{3} \leq 1$$

$$-1 \leq x \leq 1$$

$$\sin\left(\frac{n\pi}{2} + \alpha\right)$$

$\sin(\frac{n\pi}{2} + \theta)$
 $\sin \theta$
 $\cos \theta$

$$1 \leq \frac{1}{2 \cos \theta} \leq \frac{1}{3}$$

$$\sin \frac{\pi}{9} \sin \frac{2\pi}{9} \sin \frac{3\pi}{9} \sin \frac{4\pi}{9} = \frac{3}{16} \checkmark$$

$$\frac{180}{9} = 20^\circ$$

80°
40°
20°
10°

$$L.H.S = \sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$$

$$= \frac{\sqrt{3}}{2} \left[\sin 20^\circ \sin 40^\circ \sin 80^\circ \right]$$

$$= \frac{\sqrt{3}}{2} \left[\sin 40^\circ \cdot \left(\frac{-2 \sin 80^\circ \cdot \sin 20^\circ}{-2} \right) \right]$$

$$= \frac{\sqrt{3}}{4} \left[\sin 40^\circ \left(\cos 100^\circ - \cos 60^\circ \right) \right]$$

$$= -\frac{\sqrt{3}}{4} \left[\cos 100^\circ \sin 40^\circ - \cos 60^\circ \sin 40^\circ \right]$$

$$= -\frac{\sqrt{3}}{4} \left[\left(\frac{2 \cos 100^\circ \sin 40^\circ}{2} \right) - \frac{1}{2} \sin 40^\circ \right]$$

$$= -\frac{\sqrt{3}}{8} \left[\sin 140^\circ - \sin 60^\circ - \sin 40^\circ \right]$$

$$= -\frac{\sqrt{3}}{8} \left[\sin 140^\circ - \sin 40^\circ - \frac{\sqrt{3}}{2} \right]$$

$$= -\frac{\sqrt{3}}{8} \left[\sin (180^\circ - 40^\circ) - \sin 40^\circ - \frac{\sqrt{3}}{2} \right]$$

$$= -\frac{\sqrt{3}}{8} \left(\sin(\underline{\underline{180^\circ - 40^\circ}}) - \sin 40^\circ - \frac{\sqrt{3}}{2} \right)$$

$$= -\frac{\sqrt{3}}{8} \left[\cancel{\sin 40^\circ} - \cancel{\sin 40^\circ} - \frac{\sqrt{3}}{2} \right]$$

$$= \cancel{\frac{\sqrt{3}}{8}} \times \cancel{\frac{\sqrt{3}}{2}} = \frac{3}{16}$$

Express $4\sin^4 x$ in terms of an expression containing only cosines to the power 1.

$$= 4(\sin^2 x)^2$$

$$= 4\left(\frac{1 - \cos 2x}{2}\right)^2$$

$$= 1 + \cos^2(2x) - 2\cos 2x$$

$$= 1 + \frac{1 + \cos 4x}{2} - 2\cos 2x$$

$$= \frac{2 + 1 + \cos 4x - 4\cos 2x}{2} = \frac{3 - 4\cos 2x + \cos 4x}{2}$$

$$= \frac{3 - 4\cos 2x + \cos 4x}{2}$$

Example 11: Prove that: $\tan\left(\frac{\theta}{2}\right) = \pm \left(\frac{1 - \cos \theta}{\sin \theta}\right)$

Solution: We know that:

$$\tan\left(\frac{\theta}{2}\right) = \pm \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \pm \sqrt{\frac{(1 - \cos \theta)(1 - \cos \theta)}{(1 + \cos \theta)(1 - \cos \theta)}}$$

$$= \pm \sqrt{\frac{(1 - \cos \theta)^2}{1 - \cos^2 \theta}} = \pm \sqrt{\frac{(1 - \cos \theta)^2}{\sin^2 \theta}} = \pm \left(\frac{1 - \cos \theta}{\sin \theta}\right)$$

$$\frac{\sin\left(\frac{\theta}{2}\right)}{\cos\left(\frac{\theta}{2}\right)}$$

$$\sin^2 \frac{\theta}{2} = \frac{1 - \cos \theta}{2}$$

$$\sqrt{\sin^2 \frac{\theta}{2}} = \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 - \cos \theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 + \cos \theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\frac{\theta}{2} = \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}} = \frac{\sqrt{1-\cos\theta}}{\sqrt{1+\cos\theta}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$\frac{1-\cos^2\beta}{2-2\cos\beta} = \cos^2\frac{\beta}{2}$$

LHS.

$$= \frac{(1-\cos\beta)(1+\cos\beta)}{2(1-\cos\beta)}$$

$$= \frac{1+\cos\beta}{2}$$

$$= \frac{2\cos^2\left(\frac{\beta}{2}\right)}{2}$$

$$\sin 2\theta = 2\sin\theta\cos\theta$$

$$(xvii) \quad \frac{\sin\theta}{1+\cos\theta} = \tan\frac{\theta}{2}$$

$$\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}}$$

$$\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}$$

$$\begin{aligned} \cos^2\alpha &= \frac{1+\cos 2\alpha}{2} \\ \cos^2\alpha - \sin^2\alpha &= \cos 2\alpha \end{aligned}$$

$$\begin{cases} 2\cos^2\alpha - 1 \\ 1 - 2\sin^2\alpha \end{cases}$$

Roays

$$r[\sin\theta\cos\phi + \cos\theta\sin\phi]$$

13. Express the following in the form of $r \sin(\theta + \phi)$

(i) $12\sin\theta - 5\cos\theta$

(ii) $3\sin\theta + 4\cos\theta$

Let $r \cos\phi = 12 \rightarrow (1)$

$r \sin\phi = -5 \rightarrow (2)$

square (1) & (2) and add

$$r^2 \cos^2\phi + r^2 \sin^2\phi = 144 + 25$$

$$r^2 = \sqrt{169}$$

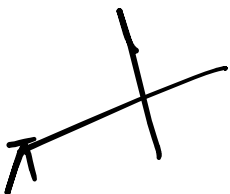
$$r = 13$$

(2) ÷ by (1),

$$\frac{r \sin\phi}{r \cos\phi} = \frac{-5}{12}$$

$$\tan\phi = \frac{-5}{12}$$

$$\phi = \tan^{-1}\left(-\frac{5}{12}\right)$$



Thus



Thus

$$12 \sin \theta - 5 \cos \theta = r \cos \phi \sin \theta + r \sin \phi \cos \theta$$

$$= r \sin(\theta + \phi)$$

$$= 13 \sin(\theta + \tan^{-1}(\frac{5}{12}))$$

$$\sec(\frac{3\pi}{2} - 62^\circ)$$

$$= -\csc 62^\circ$$

(ii) $\cot \frac{\alpha}{2} + \cot \frac{\beta}{2} + \cot \frac{\gamma}{2} = \cot \frac{\alpha}{2} \cot \frac{\beta}{2} \cot \frac{\gamma}{2}$

$$\alpha + \beta + \gamma = 180^\circ$$

$$\frac{\alpha + \beta}{2} = \frac{180^\circ - \gamma}{2}$$

$$\frac{\alpha + \beta}{2} = 90^\circ - \frac{\gamma}{2}$$

$$\cot\left(\frac{\alpha}{2} + \frac{\beta}{2}\right) = \cot\left(90^\circ - \frac{\gamma}{2}\right) \quad \text{I}$$

$$= \tan \frac{\gamma}{2}$$

$$\tan\left(\frac{\alpha}{2} + \frac{\beta}{2}\right)$$

$$\frac{1}{\frac{a}{b} \div \frac{c}{d}}$$

$$\frac{\tan \frac{\alpha}{2} + \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}} = \tan \frac{\gamma}{2}$$

$$\frac{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}}{1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2}} = \tan \frac{\gamma}{2}$$

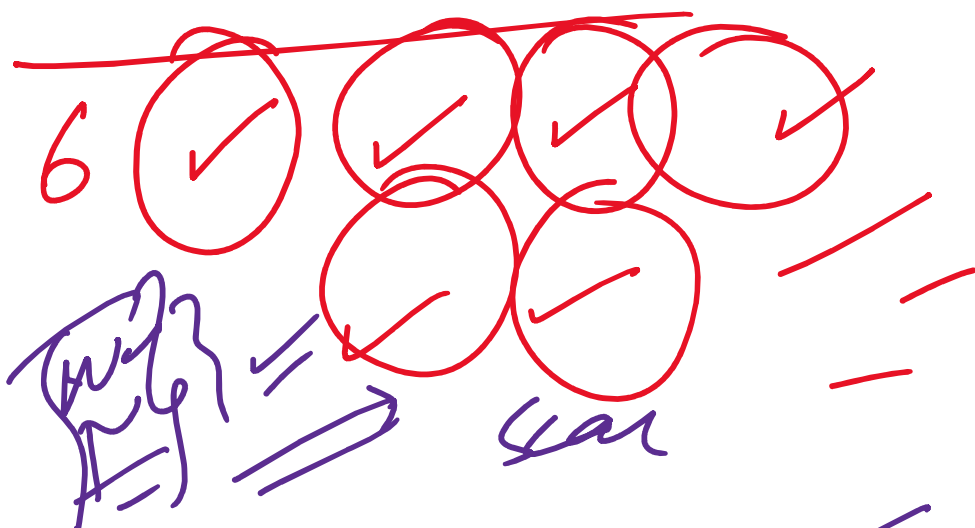
$$\left(\tan \frac{\alpha}{2} + \tan \frac{\beta}{2} \right)$$

$$1 - \tan \frac{\alpha}{2} \tan \frac{\beta}{2} = \tan \frac{\alpha}{2} \tan \frac{\gamma}{2} + \tan \frac{\beta}{2} \tan \frac{\gamma}{2}$$

$$\frac{\tan \frac{\alpha}{2} \tan \frac{\beta}{2} + \tan \frac{\alpha}{2} \tan \frac{\gamma}{2} + \tan \frac{\beta}{2} \tan \frac{\gamma}{2} = 0}{\tan \frac{\alpha}{2} \tan \frac{\beta}{2} \tan \frac{\gamma}{2}} = 1$$

$$\frac{1}{\tan \frac{\gamma}{2}} + \frac{1}{\tan \frac{\beta}{2}} + \frac{1}{\tan \frac{\alpha}{2}} = \cot \frac{\alpha}{2} \cot \frac{\beta}{2} \cot \frac{\gamma}{2}$$

$$\cot \frac{\alpha}{2} + \cot \frac{\beta}{2} + \cot \frac{\gamma}{2} = \cot \frac{\alpha}{2} \cot \frac{\beta}{2} \cot \frac{\gamma}{2}$$



dim 'x'